

Assignment#4

Computer Organization and Assembly Language Programming

Instructions

CPP file is to be submitted on portal in an uncompressed format (i.e., .cpp) before deadline.

Question

Write a C++ program that utilizes inline assembly to declare and initialize two 2x2 matrices. The objective of the program is to perform matrix multiplication on these two matrices using MMX instructions and store the resulting matrix in a third matrix.

Description

The PMADD instruction starts from a 16-bit, packed data type and generates a 32-bit packed, data type result. It multiplies all the corresponding elements generating four 32-bit results and adds the two products on the left together for one result and the two products on the right together for the other result. To complete a multiply-accumulate operation, the results would then be added to another register which is used as the accumulator.

a3	a2	a1	a0
*	*	*	*
b3	b2	b1	b0
a3*b3+a2*b2		a1*b1+a0*b0	

PMADDWD: 16b x 16b -> 32b Multiply Add

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 5+14 & 6+16 \\ 15+28 & 18+32 \end{bmatrix}$$